Professorship of Energy Management Technologies
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Assignment 2: Distribution Grid Simulation

# Meeting	Location	Content
1st Session	N3815	Introduction of assignment; work on first grids and data set
Time until 2nd Meeting	Home	Work on assignment using first two grids and data sets
2nd Session	N3815	Work on final grid; competition; presentations
Until next assignment	Home	Work on report

Background

This assignment focuses on the simulation of distribution grids and the integration of solar photovoltaic (PV) systems and heat pumps. The goal is to minimize the maximum line loadings while ensuring the grid's operational limits are not exceeded. Students will utilize the pandapower library, an open-source Python-based software for power system modeling and simulation, to model the provided distribution grids and develop a search algorithm allocating loads and generators.

The work is organized in 2 stages. Stage 1 (preparation stage) focuses on understanding distribution grid modeling, data, and task while setting up pandapower for simulation. During this stage, students will also implement their first search algorithm for optimizing the allocation of solar PV and heat pumps. Course instructors will provide guidance and feedback throughout this stage. Stage 2 (competition stage) emphasizes the application of the developed optimization algorithm in a competitive setting. Therefore, a new pandapower network will be provided in the second session. Only technical feedback will be given. Performance of the proposed allocations will be compared among the competing student groups. Grading is based on the rank in the competition, the interview being conducted in Session 2, and the report submitted before the beginning of the next assignment.

The objective of this assignment is to assess the following:

- your understanding of distribution grid simulation using pandapower,
- your capability to develop and implement search algorithms in the distribution grid context,
- your proficiency in using basic domain knowledge to create heuristics for the allocation.

The expected outcome of the assignment includes:

- a report of maximum 2 pages (excluding appendix) after the second session,
- your allocations based on the test data provided in the second session.

When preparing your development environment for this assignment, we recommend to create a new python environment and install the following packages in “one go” using pip so that it can resolve dependencies properly

- pip install “pandapower[all]” numba==0.56.4 python-igraph jupyter

Grading

The following aspects of your work will be graded:

- Description of search strategies (background and motivations, type of methods and tools employed, process of developing and refining the search algorithm, etc.)
- Rigor and creativity (originality and relevance of ideas being proposed, quality of formulation, etc.)
- Discussion (what went well, what went wrong, how the allocations could be improved - in your opinion, etc.)
- Ranking (20 bonus points first, 10 second, 5 others, 0 if no submission)
- General presentation of the assignment report (flow of information, quality of English, quality of illustrations, etc.)

Task description

Accurate distribution grid simulations to analyze renewable energy integration scenarios are essential for efficient electricity systems of the future. To gain a better understanding of these processes, you will model and simulate distribution grids and develop an optimization algorithm for allocating solar PV and heat pumps in a distribution grid. You will be given grid definitions and data sets containing different power profiles representing residential loads and generating elements, such as appliances like heat pumps, and renewable energy sources like solar PV panels. The algorithm should aim to minimize the line loading while avoiding overloading the grid. This should be done by generating optimized orders for each load and generation so that the sum of the maximum line loadings are minimal. This concept is defined programmatically in the evaluation notebook where the metric to be minimized is defined as

“res_lines.min(axis=1).sum()”

The maximum for each hour simulated is summed up. This means only the maximum line load per hour contributes to the final score that will be evaluated. Each node (bus) in the grid will have a load and a sgen allocated.

The data and various files provided for the assignment are described in detail in the Appendix.

Stage 1 – Training

Students will be provided with a figure representing a distribution grid, as well as time series data for solar PV and loads. Initially, students should familiarize themselves with the given

grid and data. Once students have a thorough understanding of the grid and data, they can proceed to develop an intelligent algorithm for allocating solar PVs and loads (including heat pumps) within the distribution grid.
Throughout the assignment, students are expected to assess the effectiveness of their proposed solution in achieving the goal of minimizing line loading. This self-evaluation process will help students refine their algorithms and improve their overall performance in the task. The students should be aware that the developed algorithms should generalize to bigger and more complex networks as well.

An evaluation file is provided for the students to check the output format of their allocations. In the competition the same file format will be used so students can test the submission and ranking procedure in the first session already.

8 files will be provided in the first session:

- **Notebook_A.ipynb:** notebook containing the first grid definition
- **GenerationData_A.csv**: file containing PV generation data for the network_A
- **LoadData_A.csv**: file containing electrical load data for the network_A
- **Notebook_B.ipynb:** notebook containing the second grid definition
- **GenerationData_B.csv**: file containing PV generation data for the network_B
- **LoadData_B.csv**: file containing electrical load data for the network_B
- **evaluation_notebook_students.csv**: file for structure of evaluation, to check if your allocations have the right format
- **utils.py:** auxiliary file

Stage 2 - Competition

In the competition stage, the student groups receive two final datasets and a new, more complex grid definition to optimize the ordering and are asked to generate allocations to compete against each other.

Like in the preparation stage please upload your final allocations as a .csv-File on Moodle.

The uploaded file has the name (in python f-string fashion):

f'Competition_2_Group_{Group_letter}.csv'

The name of the columns are 'generation_order' and 'load_order'.

These data will be evaluated by the tutors and the rankings will be made.

3 files will be provided in the second session:

- **Notebook_C.ipynb:** notebook containing the third grid definition
- **GenerationData_C.csv**: file containing PV generation data for the network_C
- **LoadData_C.csv**: file containing electrical load data for the network_C

Structure and contents of the report to be delivered

The report for the assignment should include:

- A description of main motivation behind your search strategy.
- An evaluation of different allocations of PVs and loads.
- A few plots of some line loadings and the power drawn from the external grid
- Some proposals for how you would improve your search strategy.
- The code as .ipynb or .py files

The report files and the code must be uploaded as one .zip file to Moodle before the given due date.

Naming of Submissions

Each submission must be named according to the following pattern:

- Competition: f"Competition_2_Group_{Group_letter}.csv“
- Presentation: f"Presentation_2_Group_{Group_letter}.pdf“
- Report and Code: f"Report_Code_2_Group_{Group_letter}.zip“

As an example for the group submissions of Group E:

- “Competition_2_Group_E.csv”
- “Presentation_2_Group_E.pdf”
- "Report_Code_2_Group_E.zip“

Suggested readings:

- https://www.pandapower.org/start/
- https://pandapower.readthedocs.io/

Appendix: Short description of data and various files

The various files include:

GenerationData_{A,B,C}.csv

This file gives you the generation data.

Session 1: A, B

Session 2: C

Columns: index: time step 0-23, 0-n (households): generation in Megawatt

The single rows represent for one time step all the generation powers for all nodes.

LoadData_{A,B,C}.csv

This file gives you the electrical load data.

Session 1: A, B

Session 2: C

Columns: index: time steps 0-23, 0-n (households): electrical load in Megawatt

The single rows represent for one time step all the electrical loads for all nodes.